Edge Computing and Intelligence: A Survey

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Abstract

Edge computing is a rapidly evolving field that has the potential to revolutionize various industries and applications by providing low-latency, high-bandwidth, and real-time processing capabilities. This survey provides a comprehensive overview of the current state of research in edge computing, including its definition, importance, and current state of research, as well as its applications in IoT, smart cities, industrial automation, and healthcare. The key findings of this survey highlight the importance of edge computing in reducing latency, improving real-time processing, and enhancing overall system efficiency. Future directions for edge computing research include exploring new applications and services, developing more efficient edge computing architectures and algorithms, and addressing the challenges and limitations identified in this survey. With its potential to enable a wide range of applications, from IoT and smart cities to industrial automation and healthcare, edge computing is an exciting and rapidly evolving field with a wide range of opportunities for future research and development.

1 Introduction

1.1 Introduction to Edge Computing

Edge computing is a distributed computing paradigm that brings computation and data storage closer to the source of the data, reducing latency and improving real-time processing [1, 2, 3, 4, 5, 6, 7, 8, 9]. This paradigm enables data processing and analysis at the edge of the network, thereby reducing latency and improving real-time processing [10]. By moving the computational load towards the edge of the network, edge computing can harness untapped computational capabilities in edge nodes, reducing latency and improving real-time processing [11, 12, 13, 14, 15, 16]. The importance of edge computing lies in its ability to provide reduced latency, increased bandwidth, and support for real-time applications [17]. Edge computing has several benefits, including low-latency, energy-efficiency, privacy protection, reduced bandwidth consumption, on-premises, and location awareness [11, 12, 13, 14, 15, 16]. Furthermore, edge computing is crucial for Edge Information Systems (EIS) for intelligent Internet of Vehicles (IoV), edge caching, edge computing, and edge AI [18]. As noted by [19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88], edge computing is a critical component of modern computing systems, and its importance will continue to grow as the demand for real-time processing and analysis of data increases.

1.2 Current State of Research in Edge Computing

The current state of research in edge computing is rapidly evolving, with numerous initiatives and advancements being made in recent years [89, 90, 91, 92, 93, 94, 95, 96]. The field of edge computing is experiencing significant growth, with a focus on microservices and their applications in edge computing [90]. The survey by [91] organizes the current methods and research into several fields, including mobile edge computing, fog computing, cloudlet, and edge caching. Moreover, the evolution of next-generation IoT networks towards the edge is driven by the introduction of intelligent IoT

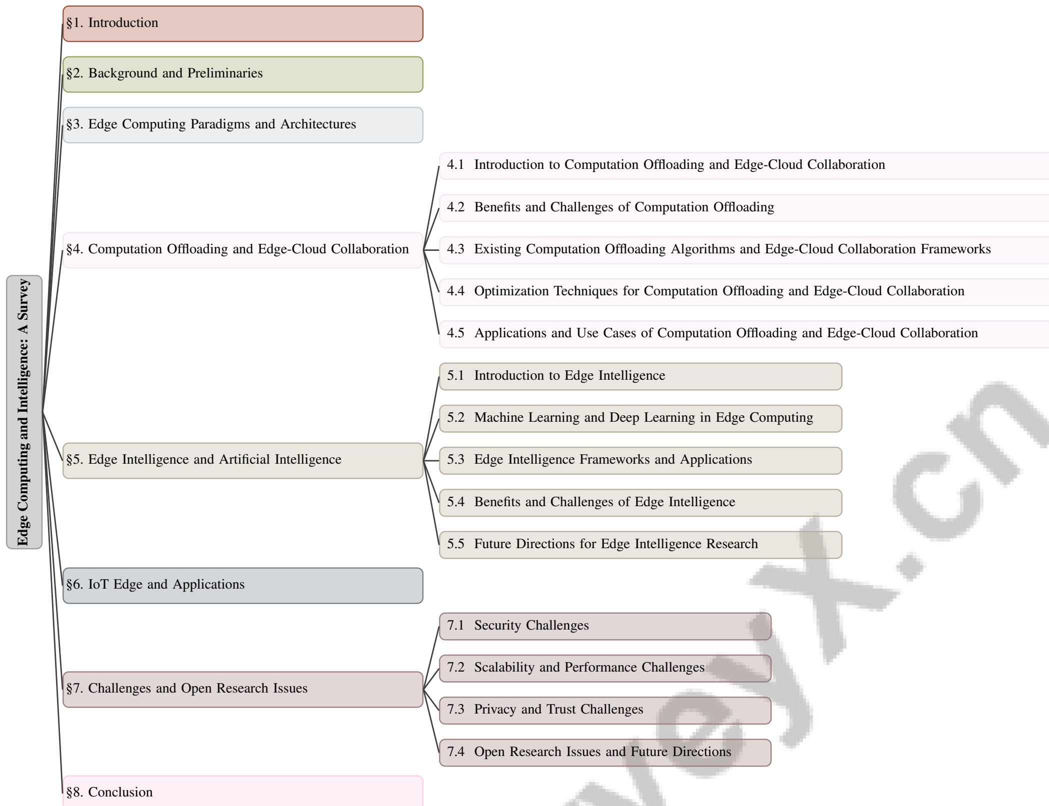

Figure 1: chapter structure

environments, which characterizes the energy efficiency of complex systems, optimizes the allocation of application components to devices, and ensures trustworthiness in decentralized and heterogeneous scenarios [93]. The integration of terrestrial and non-terrestrial networks, including unmanned aerial vehicles (UAVs), high-altitude platforms (HAPs), and low-earth orbit (LEO) satellites, is also being explored to provide ubiquitous connectivity and enhanced capacity [96]. As noted by [89, 90, 91, 92, 93, 94, 95, 96], edge computing has the potential to revolutionize various industries and applications, and its importance will continue to grow as the demand for real-time processing and analysis of data increases.

1.3 Importance of Edge Computing

Edge computing has numerous benefits, including improved system efficiency and reduced latency, making it a crucial component of modern computing systems [97, 10, 17]. The importance of edge computing lies in its ability to provide reduced latency, increased bandwidth, and support for real-time applications [10]. By moving the computational load towards the edge of the network, edge computing can harness untapped computational capabilities in edge nodes, reducing latency and improving real-time processing [11, 12, 13, 14, 15, 16]. Edge computing offers advantages such as reduced latency, improved reliability, and enhanced security, highlighting its importance in improving system efficiency [97]. The benefits of edge computing are also emphasized by its ability to enable reduced latency, energy efficiency, and real-time applications, making it an attractive solution for a wide range of applications [98]. As noted by [11, 12, 13, 14, 15, 16], edge computing is essential for improving system efficiency, reducing latency, and enhancing overall system performance, and its importance will continue to grow as the demand for real-time processing and analysis of data increases.

1.4 Structure of the Survey

This survey is organized into several sections, each focusing on a specific aspect of edge computing [89]. The introduction provides an overview of the topic, including the definition, importance, and

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current state of research in edge computing. The background and preliminaries section delves into the core concepts involved in edge computing, including limitations of traditional cloud computing, the need for edge computing, benefits and challenges of edge computing, and edge computing architectures and key technologies. The section on edge computing paradigms and architectures discusses the different edge computing paradigms, including multi-access edge computing, fog computing, and cloudlet, and compares their advantages and disadvantages. The computation offloading and edge-cloud collaboration section explores the concept of computation offloading, its benefits and challenges, and existing computation offloading algorithms and edge-cloud collaboration frameworks. The edge intelligence and artificial intelligence section discusses the concept of edge intelligence, its relationship to artificial intelligence, and the use of machine learning and deep learning in edge computing. The IoT edge and applications section examines the application of edge computing in IoT scenarios, including smart cities, industrial automation, and healthcare. The challenges and open research issues section discusses the challenges and open research issues in edge computing, including security, privacy, and scalability. Finally, the conclusion summarizes the key findings of the survey and highlights the importance of edge computing in reducing latency, improving real-time processing, and enhancing overall system efficiency. As noted by [11, 12, 13, 14, 15, 16], this survey aims to provide a comprehensive overview of the current state of research in edge computing, its benefits and challenges, and its applications in various domains.The following sections are organized as shown in Figure 1.

2 Background and Preliminaries

2.1 Limitations of Traditional Cloud Computing

Traditional cloud computing is hindered by several limitations, including latency, low bandwidth, and security concerns, particularly in IoT, smart cities, and healthcare [17, 1, 89, 10, 99, 100, 91]. The need for low-latency processing, context-awareness, mobility support, and scalability are also key limitations of traditional cloud computing [89, 100]. Furthermore, traditional cloud computing is limited by the lack of resources for failover, security concerns, and the need for standardization [10, 99]. The high cost and potential interruptions of data transmission from the cloud to edge servers are also significant limitations [99, 101]. In Mobile Edge Computing (MEC), the key obstacles include the need for seamless integration of wireless communications and mobile computing, the management of radio-and-computational resources, and the provision of low-latency and high-reliability services [100, 93, 96].

The limitations of traditional cloud computing have led to the development of edge computing as a means to address these limitations [17, 1, 102, 46, 57, 61]. The resource-constrained nature of the edge, which can result in higher end-to-end latency, especially at higher utilizations, when compared to cloud data centers, is another significant limitation [103, 101]. The lack of a global perimeter in edge paradigms, which makes it difficult to protect against security threats, and the decentralized and distributed nature of these paradigms, as well as the existence of mobile devices and user-owned edge data centers, adds to the complexity of securing these environments [101, 93].

2.2 Need for Edge Computing

The need for edge computing arises from the limitations of traditional cloud computing in terms of latency, bandwidth, and security, as well as the increasing demand for real-time processing and analysis of data [31, 102, 104, 20]. According to [31, 102], the need for edge computing is driven by the requirement for ultra-reliable low-latency communication in mobile edge computing systems and the need for a computing paradigm that can handle large amounts of data generated by IoT devices at the edge of the network with low latency and high reliability.

The importance of edge computing lies in its ability to provide reduced latency, increased bandwidth, and support for real-time applications, such as virtual reality, vehicle-to-everything, and edge artificial intelligence, by leveraging computational resources closer to data sources, thereby enabling faster responses, improving data security, and enhancing the quality of user experience [17, 97, 104, 20]. Therefore, the need for edge computing is driven by the limitations of traditional cloud computing, including its inability to handle the massive quantities of data generated by IoT devices, smartphones, and other edge devices, as well as the increasing demand for real-time processing and analysis of data [17, 104, 105, 89].

2.3 Benefits and Challenges of Edge Computing

Edge computing has numerous benefits, including improved system efficiency and reduced latency, making it a crucial component of modern computing systems [102, 38, 54, 55]. The benefits of edge computing include reduced latency, increased bandwidth, and support for real-time applications [102, 54]. According to [102, 54], edge computing offers advantages such as low latency and high reliability, support for real-time processing and decision-making, and potential to reduce bandwidth consumption and improve quality of experience.

However, edge computing also poses challenges, such as security and scalability [54, 85]. The challenges of edge computing include the heterogeneity of edge resource capacities and the inconsistence of storage and computation capacities of edge nodes [85, 100]. Additionally, edge computing relies on various networking technologies, such as 5G and Wi-Fi, which provide high-speed and low-latency connectivity for edge devices [100, 96].

2.4 Edge Computing Architectures and Key Technologies

Edge computing architectures and key technologies are crucial components of modern computing systems, enabling data processing and analysis at the edge of the network, reducing latency and improving real-time processing [106, 107, 108, 89]. According to [106, 107], IoTSim-Edge is a simulation framework that models the behavior of heterogeneous IoT and edge computing infrastructure, allowing users to test their infrastructure and framework in an easy and configurable manner.

Key technologies in edge computing include edge computing platforms, such as EdgeX Foundry and Open Edge Computing, which provide a framework for developing and deploying edge computing applications that can harness the abundant resources at the network's edge, reduce communication latency, and augment privacy [107, 108, 89]. Other key technologies include containerization and orchestration tools, such as Docker and Kubernetes, which enable efficient deployment and management of edge computing applications [54, 10].

The components of edge computing architectures include edge devices, edge nodes, and cloud servers, which work together to provide a scalable and flexible computing infrastructure [10, 17, 89]. Edge devices, such as IoT devices and smartphones, generate data that needs to be processed and analyzed [17, 89]. Edge nodes, such as edge servers and gateways, provide processing and storage capabilities for data generated by edge devices [89, 54]. Cloud servers and data centers provide additional processing and storage capabilities for data that requires more complex analysis or longer-term storage [10, 102].

The functionalities of edge computing architectures include data processing, data storage, and data analysis [102, 54]. Edge computing architectures can process data in real-time, reducing latency and improving system efficiency [54, 55]. They can also store data locally, reducing the need for data transmission to the cloud, which not only improves system security by minimizing the risk of data breaches and unauthorized access, but also enables faster response times, lowers latency, and decreases communication overhead [43, 109, 104, 110].

3 Edge Computing Paradigms and Architectures

The field of edge computing has experienced significant growth in recent years, driven by the increasing demand for low-latency, high-bandwidth, and real-time processing capabilities, as noted by [31, 102]. This growth has led to the emergence of various edge computing paradigms and architectures, each with its own strengths and weaknesses, as discussed in [91, 93, 95, 28, 37]. According to [102, 48, 57, 111], edge computing paradigms are designed to provide low-latency and high-bandwidth services for applications that require real-time processing and analysis of data, and they have the potential to support a wide range of applications, including IoT, smart cities, healthcare, and industrial automation. As illustrated in Figure 2, the hierarchical structure of edge computing paradigms and architectures can be categorized into decentralized cloud and low latency computing, fog computing, and cloudlet computing, highlighting key characteristics such as low-latency, high-bandwidth, and real-time processing. This figure also provides a comparison of different edge computing paradigms and architectures, including vMEC, iSapiens, and SmartEdge, and identifies primary challenges and open research issues in edge computing, such as management of distributed and heterogeneous resources, lack of resources for failover, and meeting real-time deadlines. The visualization of these concepts in Figure 2 enhances our understanding of the complex relationships between various edge computing paradigms and architectures, and underscores the need for further research into addressing the challenges and open issues in this field. By examining the current state of edge computing paradigms and architectures, as well as the challenges and open research issues, researchers and practitioners can work towards developing more efficient, scalable, and reliable edge computing systems that can support a wide range of applications and services.

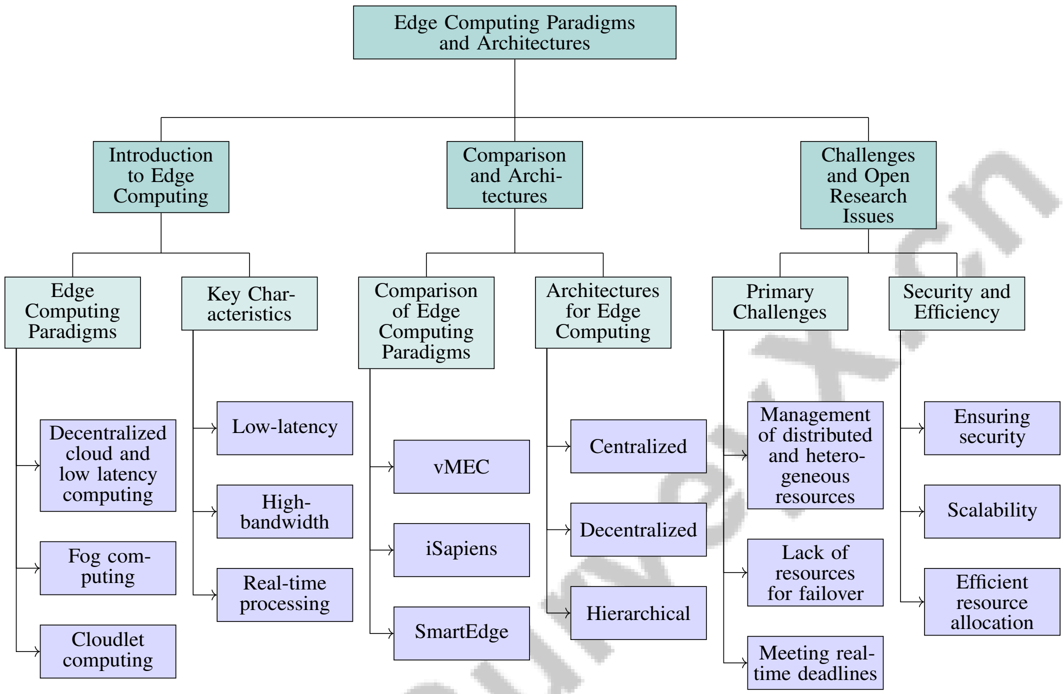

Figure 2: This figure illustrates the hierarchical structure of edge computing paradigms and architectures, including introduction to edge computing, comparison and architectures, and challenges and open research issues. The diagram categorizes edge computing paradigms into decentralized cloud and low latency computing, fog computing, and cloudlet computing, and highlights key characteristics such as low-latency, high-bandwidth, and real-time processing. It also compares different edge computing paradigms and architectures, including vMEC, iSapiens, and SmartEdge, and identifies primary challenges and open research issues in edge computing, such as management of distributed and heterogeneous resources, lack of resources for failover, and meeting real-time deadlines.

3.1 Introduction to Edge Computing Paradigms

Edge computing paradigms are diverse and include various architectures, such as decentralized cloud and low latency computing, fog computing, and cloudlet computing, which involve leveraging edge nodes to perform computations and reduce latency, as noted by [91, 93, 95]. The survey by [95] introduces a novel perspective on fog and edge computing, highlighting their potential to support a wide range of applications and services. Moreover, [28] provides a novel viewpoint on the architecture of edge computing paradigms, describing the hierarchical structure of edge intelligence, involving multiple tiers, including edge nodes, fog nodes, and cloudlets. This hierarchical structure is further illustrated in Figure 3, which depicts the organization of edge computing paradigms, including fog computing, edge architecture, and edge applications, providing a visual representation of the concepts discussed in relevant research papers, such as those by [37], which introduces a novel viewpoint on IoT architecture, integrating SDN and Edge Computing to support distributed controllers and connected switches built on the OpenFlow framework. The combination of these architectural elements enables the support of distributed applications and services, underscoring the complexity and versatility of edge computing paradigms.

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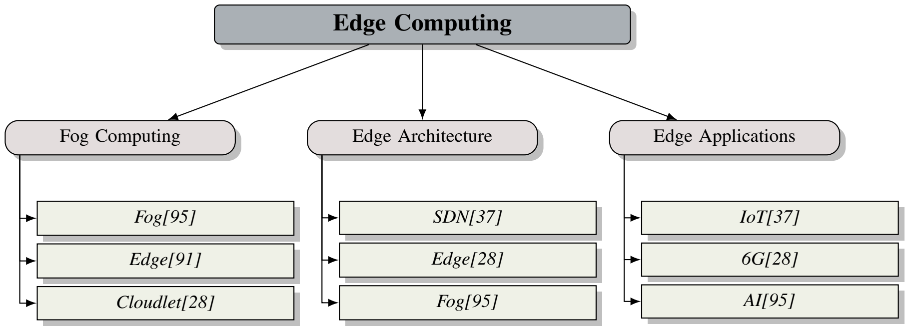

Figure 3: This figure shows the hierarchical structure of edge computing paradigms, including fog computing, edge architecture, and edge applications, with references to relevant research papers.

3.2 Comparison of Edge Computing Paradigms

The comparison of edge computing paradigms is crucial in understanding their advantages and limitations, and how they can be applied in various scenarios, as discussed in [1, 17, 112]. According to [17], different edge computing techniques and models, such as vMEC, iSapiens, and SmartEdge, have their advantages and limitations in various applications. For instance, vMEC is suitable for applications that require low latency and high bandwidth, while iSapiens is more suitable for applications that require intelligent decision-making and real-time processing. Moreover, [112] compares different fog computing platforms, including private, public, and open-source platforms, highlighting their strengths and weaknesses.

3.3 Architectures for Edge Computing

The architectures for edge computing are diverse and include various components and functionalities, such as data flow, control, and tenancy architectures, as noted by [113, 112]. According to [112], the stages of fog computing architecture include data collection, processing, and analysis at the edge of the network, with the possibility of offloading tasks to the cloud. The survey by [95] identifies three main types of architectures for fog and edge computing: centralized, decentralized, and hierarchical, each with its own strengths and weaknesses. The components of edge computing architectures include edge devices, edge nodes, and cloud servers, which work together to provide a scalable and flexible computing infrastructure, as discussed in [10, 17, 89]. Edge computing architectures can process data in real-time, reducing latency and improving system efficiency, and they can also store data locally on edge devices, reducing the need for data transmission to the cloud, which not only improves system security by minimizing the risk of data breaches during transmission but also enables faster response times, lowers latency, and enhances overall energy efficiency, as noted by [109, 43, 17, 104, 110].

3.4 Challenges and Open Research Issues in Edge Computing Paradigms and Architectures

The challenges and open research issues in edge computing paradigms and architectures are numerous and complex, requiring careful consideration and innovative solutions, as discussed in [112, 10, 103]. One of the primary challenges is the management of distributed and heterogeneous resources, which is essential for ensuring efficient resource allocation, scalability, and security in edge computing environments. According to [10], the challenges in edge computing include the lack of resources for failover, meeting real-time deadlines, authentication and physical security, multi-tenancy, and standardization. The need for dynamic edge resource allocation techniques is critical to avoid or mitigate the edge performance inversion problem, which can significantly impact the efficiency and effectiveness of edge computing systems, as noted by [103]. Additionally, the challenges in edge computing paradigms and architectures include ensuring security, scalability, and efficient resource allocation, as discussed in [112, 10].

4 Computation Offloading and Edge-Cloud Collaboration

4.1 Introduction to Computation Offloading and Edge-Cloud Collaboration

Computation offloading and edge-cloud collaboration are vital components of edge computing, enabling efficient data processing and analysis at the network edge, as discussed in [114, 97, 89]. The integration of IoT with cloud, fog, and edge computing involves data collection, processing, and analysis, utilizing various protocols and technologies, such as RFID, Wi-Fi, and 5G networks [114]. A framework for understanding the complex Edge Computing problem space is provided in [89], distinguishing between technical layers and hierarchy levels. Primary challenges in computation offloading include task partitioning, quality-of-service, and security [105, 115].

Computation offloading and edge-cloud collaboration can reuse previously executed tasks to achieve lower response times and efficiently utilize edge resources [116]. Additionally, they can alleviate latency, communication costs, and privacy concerns in deploying machine learning models on edge devices [117]. Researchers face challenges such as dynamicity, low-latency, and high-throughput requirements, as well as complexity in managing Microservices [90]. The benefits of computation offloading and edge-cloud collaboration include improved system efficiency, reduced latency, and enhanced security and privacy [17, 1].

The hierarchical structure of computation offloading and edge-cloud collaboration is illustrated in Figure 4, which categorizes edge computing into MEC, Fog Computing, and Cloudlet. The computation offloading category is further divided into task offloading, resource allocation, and energy efficiency, while edge-cloud collaboration encompasses latency minimization, security, and privacy. This hierarchical structure highlights the intricate relationships between these components and underscores the importance of optimizing each category to achieve efficient computation offloading and edge-cloud collaboration.

Computation offloading optimization in edge computing-supported IoT is critical, where IoT devices can offload tasks to edge nodes to optimize problems such as delay and connectivity [118]. Mobile devices can offload processing tasks to the cloud or process them locally, as discussed in [119]. The technical aspects of MEC-IoT integration, including computation offloading, are discussed in [120], highlighting the benefits of MEC in improving scalability and reducing latency in IoT applications. Computation offloading and edge-cloud collaboration are applied in various areas, such as drone follow-me services, remote video analytics, and intelligent IoV [121, 18].

Energy-efficient dynamic computation offloading in edge-computing-aided beyond 5G networks is another important aspect, as discussed in [122]. A system model combining mobile edge computing and spectrum sharing for 5G communications is proposed in [123], related to computation offloading and edge-cloud collaboration for IoT networks. Computation offloading and edge-cloud collaboration can reduce latency in MEC networks, as discussed in [124]. The edge data dissemination problem, involving data dissemination from the cloud to multiple target edge servers, is explored in [99].

MEC enables computation-intensive and latency-critical applications at resource-limited mobile devices by providing IT and cloud-computing capabilities within the RAN [100]. A survey discussing key technologies that enable mobile edge networks, including cloud technology, software-defined networking, and network function virtualization, is presented in [91]. Energy-efficient computation offloading and edge-cloud collaboration can be applied to AR applications, reducing mobile energy consumption [125]. The importance of considering the energy-performance tradeoff in the design of iiote and the need for a holistic approach to integrating enabling technologies are critical aspects of computation offloading and edge-cloud collaboration [93].

Computation offloading in mobile edge computing, cooperative node-based offloading, and joint optimization of duration and resource allocation are discussed in [94]. The integration of Blockchain technology with MEC for secure offloading is explored in [126]. The concept of computation offloading and edge-cloud collaboration is related to edge intelligence, enabling new services and applications, as discussed in [28]. Computation offloading strategies in integrated terrestrial and non-terrestrial networks, including edge, cloud, and hybrid approaches, are essential for efficient data processing and analysis [96]. A new three-tier IoT architecture, integrating SDN and Edge Computing, is proposed in [37].

In the context of Mobile Edge Computing (MEC) environment, computation offloading is crucial for complex applications represented as workflows with task dependencies, where the goal is to minimize end-device's energy consumption under given deadline constraints, as discussed in [127]. Computation offloading and edge-cloud collaboration are critical in mobile edge computing, where services should be dynamically migrated among multiple edges to maintain service performance [46]. The concept of computation offloading and edge-cloud collaboration is applied in MEC-enabled Metaverse, optimizing transmission data size and user-server association to maximize users' utility, as discussed in [48].

A joint computation and communication cooperation approach is proposed in [128] to improve MEC performance. The problem of computation offloading and edge-cloud collaboration is addressed in [52], where task assignment, computing, and transmission resource allocation are optimized. Computation offloading in Mobile Edge Computing (MEC) is discussed in [129], particularly for efficient distributed resource management, using game theory and reinforcement learning solutions. The concept of computation offloading is introduced in [57], where edge-cloud collaboration is discussed as a means to improve system efficiency and reduce latency.

Computation offloading and edge-cloud collaboration can help overcome the limitations of traditional cloud computing, but also require a comprehensive and trusted architecture to fulfill military requirements, as noted by [72]. Introduction to computation offloading and edge-cloud collaboration, including the concept of offloading computationally intensive workloads to the MEC server, is discussed in [130]. The joint optimization of service placement and request routing in MEC-enabled multi-cell networks is a key problem to be addressed, as highlighted by [75].

The proposed idea is to use Fog computing to improve the execution of big-data analytics and cyber security applications, as discussed in [131]. The concept of computation offloading and edge-cloud collaboration, optimizing task processing performance under long-term cost constraints, is introduced in [132]. Computation offloading and edge-cloud collaboration can be applied to mobile edge computing to minimize service response time and outsourcing traffic to central clouds, as noted by [85].

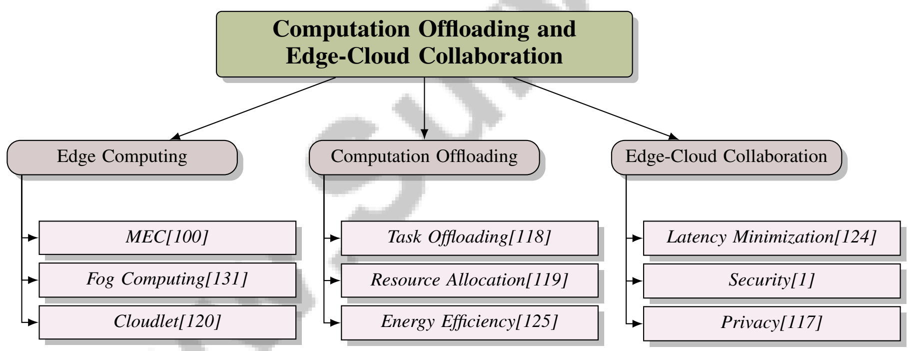

Figure 4: This figure shows the hierarchical structure of computation offloading and edge-cloud collaboration, including edge computing, computation offloading, and edge-cloud collaboration. The edge computing category includes MEC, Fog Computing, and Cloudlet. The computation offloading category includes task offloading, resource allocation, and energy efficiency. The edge-cloud collaboration category includes latency minimization, security, and privacy.

4.2 Benefits and Challenges of Computation Offloading

Computation offloading is a critical component of edge computing, enabling efficient data processing and analysis at the network edge. The benefits of computation offloading include decreased completion time, optimized resource utilization, and high computation gain with high accuracy and correctness, as discussed in [116, 132]. Computation offloading can reduce latency, improve bandwidth utilization, and enhance security and privacy, as noted by various studies.

However, there are also challenges in computation offloading, including the need for efficient coordination mechanisms, handling of mobility and intermittent connectivity, and ensuring security and reliability. The complexity of proposed schemes, such as optimization problems in cloud and edge computing, can pose challenges in computation offloading, as discussed in [132]. The heterogeneity of edge resource capacities and the inconsistency of storage and computation capacities of edge nodes can affect computation offloading performance, as noted by [85].

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The proposed algorithm can schedule tasks based on deadlines and required CPU cycles, ensuring that tasks with earlier deadlines and smaller CPU cycle requirements are prioritized, as discussed in [133]. Computation offloading can reduce service response time and core network traffic, which is essential for applications requiring low latency and high bandwidth, as noted by [85].

4.3 Existing Computation Offloading Algorithms and Edge-Cloud Collaboration Frameworks

Existing computation offloading algorithms and edge-cloud collaboration frameworks are crucial components of edge computing, enabling efficient data processing and analysis at the network edge. The proposed spatiotemporal non-uniformity-aware online task scheduling scheme is an existing computation offloading algorithm that considers the spatiotemporal non-uniformity of task arrivals and edge server capacities, as discussed in [132]. The ICE algorithm, proposed in [85], jointly optimizes service caching and workload scheduling policies to minimize service response time and outsourcing traffic to central clouds.

The Optimal Job Scheduling Algorithm (OJSA), discussed in [133], determines the optimal task order for a given set of tasks, enabling users to make informed decisions for offloading tasks based on server information. The Decoupled Access for Mobile Edge Computing (DA-MEC) method, proposed in [134], uses stochastic geometry tools to analyze the performance of the decoupled access scheme and considers backhaul capacity limitations. The Graph4Edge-Nonlinear algorithm, proposed in [127], is a novel graph-based computation offloading strategy that handles nonlinear workflow structures and finds the optimal offloading decision with minimum energy consumption under deadline constraints.

The D-VNFP algorithm, proposed in [135], deploys user requests to satellite edge servers and a cloud data center during a time slot while minimizing network bandwidth usage and user end-to-end delay. The Latency Minimization Algorithm (LMA), proposed in [52], jointly coordinates task assignment, computing, and transmission resource allocation to minimize system latency in the HetMEC network. Game theory and reinforcement learning approaches for computation offloading and resource management in MEC are discussed in [129].

The proposed Lyapunov optimization-based dynamic computation offloading (LODCO) algorithm, discussed in [130], is another existing computation offloading algorithm. The concept of computation offloading and edge-cloud collaboration is applied in various areas, such as MEC-enabled Metaverse, where optimizing transmission data size and user-server association can maximize users' utility, as discussed in [48]. A joint computation and communication cooperation approach is proposed in [128] to improve MEC performance.

4.4 Optimization Techniques for Computation Offloading and Edge-Cloud Collaboration

Optimization techniques for computation offloading and edge-cloud collaboration are crucial in achieving efficient and effective edge computing systems. The advantages of edge cloud offloading include low latency, energy efficiency, and support for resource-intensive applications, as discussed in [115]. To achieve these advantages, optimization techniques such as minimizing task completion time by increasing the possibility of reusing previously executed tasks and minimizing communication cost by reducing content traversing the core network can be employed, as noted by [116].

Optimization techniques can be categorized into centralized and distributed methods, as discussed in [136]. Centralized methods involve a central controller making decisions about computation offloading and resource allocation, while distributed methods involve multiple controllers making decisions independently. The performance of these methods can be evaluated in terms of latency, energy consumption, cost, utility, profit, and resource utilization, as noted by [136].

The principle of FL-DRL can be applied to computation offloading and energy allocation, as discussed in [118]. FL-DRL uses deep reinforcement learning (DRL) to make decisions about computation offloading and energy allocation and federated learning (FL) to train the DRL agents in a distributed manner. Virtual queues can be used to handle long-term constraints, and a drift-plus-penalty function can be used to minimize the weighted sum of energy consumption and delay, as noted by [122].

The use of blockchain technology can provide secure offloading, and different schemes can be evaluated in terms of their potential in achieving secure offloading, as discussed in [126]. The Lyapunov optimization framework can be used to decouple the long-term optimization problem into per-frame deterministic problems, and a model-free deep reinforcement learning algorithm can be applied to optimize computation offloading and edge-cloud collaboration, as noted by [137].

A graph-based approach can be used to optimize computation offloading, and the Dijkstra algorithm can be used to find the shortest path in the Energy Transmission Graph (ETG), as discussed in [127]. Lyapunov optimization and graph-based models can also be used to optimize computation offloading and edge-cloud collaboration, as noted by [132].

4.5 Applications and Use Cases of Computation Offloading and Edge-Cloud Collaboration

Computation offloading and edge-cloud collaboration have numerous applications and use cases in various domains, including smart health, smart home, smart city, and IoT systems, as discussed in [115]. The application of Graph4Edge-Nonlinear in MEC-based UAV delivery systems is another example, evaluated through real-world case studies and simulation experiments on the FogWorkflowSim platform, as noted by [127].

In smart health, computation offloading and edge-cloud collaboration can process and analyze medical data in real-time, enabling faster diagnosis and treatment, as discussed in [115]. In smart home, they can control and automate devices, improving energy efficiency and convenience, as noted by [115]. In smart city, computation offloading and edge-cloud collaboration can manage and analyze traffic, energy, and other urban systems, improving public safety and quality of life, as discussed in [115].

Computation offloading and edge-cloud collaboration can be used in IoT systems to process and analyze data from various sensors and devices, enabling real-time monitoring and decision-making, as noted by [115]. The use of Graph4Edge-Nonlinear in MEC-based UAV delivery systems is another example of the application of computation offloading and edge-cloud collaboration in IoT systems, as discussed in [127].

In addition, computation offloading and edge-cloud collaboration can be used in various other applications, such as augmented reality, virtual reality, and gaming, where low latency and high bandwidth are critical, as noted by [127]. The evaluation of Graph4Edge-Nonlinear in MEC-based UAV delivery systems through real-world case studies and simulation experiments on the FogWorkflowSim platform demonstrates the effectiveness of computation offloading and edge-cloud collaboration in these applications, as discussed in [127].

5 Edge Intelligence and Artificial Intelligence

This section provides an overview of the intersection of edge intelligence and artificial intelligence, highlighting the critical role of edge intelligence in enabling efficient processing and analysis of data at the edge of the network. With the increasing demand for real-time processing and analysis of data, edge intelligence has become a crucial component of edge computing, leveraging machine learning and deep learning algorithms to provide low-latency and high-bandwidth applications. The following subsections will delve into the key aspects of edge intelligence, including its introduction, the role of machine learning and deep learning, edge intelligence frameworks and applications, benefits and challenges, and future research directions.

5.1 Introduction to Edge Intelligence

Edge intelligence, a rapidly evolving field that converges edge computing and artificial intelligence, is a critical component of edge computing, enabling the efficient processing, analysis, and real-time decision-making of the vast amounts of data generated by billions of IoT devices at the edge of the network, thereby unlocking new opportunities for AI-driven applications and services, such as intelligent surveillance, personalized recommendations, and autonomous systems, while addressing key challenges related to data confidentiality, integrity, and scalability. This convergence is illustrated in Figure 5, which depicts the hierarchical structure of Edge Intelligence, including its definition, applications, and challenges, providing a comprehensive overview of the field and its relevance to edge computing.

The relationship between edge intelligence and artificial intelligence is critical, as edge intelligence relies on the use of machine learning and deep learning algorithms to process and analyze data in real-time at the edge of the network, leveraging the benefits of edge computing to reduce latency, improve efficiency, and enhance data confidentiality and integrity, while also enabling the development of new applications such as immersive video conferencing, augmented/virtual reality, and autonomous vehicles, and paving the way for innovative solutions to address the challenges of scalability, security, and resource constraints in edge AI systems. As shown in Figure 5, the applications of Edge Intelligence are diverse and rapidly expanding, with potential uses in various fields, including edge-based video analytics, mobile edge computing, and IoT-Fog-Cloud architectures.

Edge intelligence focuses on deploying machine learning systems on edge-devices to improve privacy, security, and system response times. The use of machine learning and deep learning in edge computing enables the efficient processing and analysis of data, which is essential for applications that require real-time decision-making. Moreover, edge intelligence can be applied to various applications, including edge-based video analytics, enabling the use of machine learning and deep learning techniques for real-time video processing and analysis. The proposed IoT-Fog-Cloud architecture can effectively implement big-data analytics and cyber security applications, and the use of Fog technology can tackle the drawbacks of Cloud systems. IoTSim-Edge can capture the complete behavior of IoT and edge computing infrastructure development and deployment, which is related to edge intelligence [106].

In addition, edge intelligence can be applied to mobile edge computing to optimize service caching and workload scheduling policies, and to minimize the service response time and the outsourcing traffic to central clouds [85]. The importance of edge intelligence in edge computing lies in its ability to provide real-time processing and analysis of data, which is essential for applications that require low latency and high bandwidth. The relationship between edge intelligence and artificial intelligence is also critical, as edge intelligence relies on the use of machine learning and deep learning algorithms to process and analyze data at the edge. As noted by [133], the proposed algorithm is related to edge intelligence, as it is designed to maximize the number of tasks completed before their predefined deadlines in mobile edge computing (MEC) systems. The algorithm can be used to make informed decisions for offloading tasks based on the information provided by servers. Furthermore, [16] highlights the concept of edge intelligence, its importance in edge computing, and the relationship between edge intelligence and artificial intelligence.

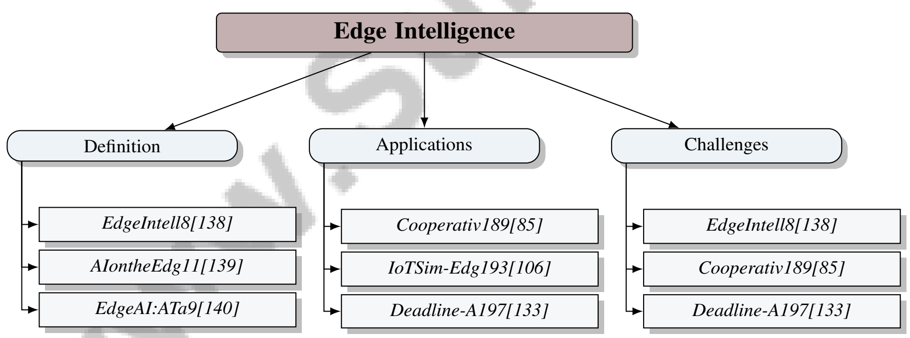

Figure 5: This figure shows the hierarchical structure of Edge Intelligence, including its definition, applications, and challenges, with references to relevant papers.

5.2 Machine Learning and Deep Learning in Edge Computing

Machine learning and deep learning are crucial components of edge computing, enabling the efficient processing and analysis of data at the edge of the network . According to [138], the survey organizes the current research efforts in Edge Intelligence into several fields and stages, including Topology, Content, and Service for Edge Computing, and Model Adaptation, Framework Design, and Processor Acceleration for AI on Edge. This highlights the importance of machine learning and deep learning in edge computing, as they enable the development of intelligent edge devices and applications.

The use of machine learning and deep learning in edge computing can improve system efficiency and reduce latency [117]. For instance, machine learning algorithms can be used to optimize computation offloading and energy allocation in edge computing, as demonstrated by [118]. Deep learning techniques, such as deep reinforcement learning, can also be used to make decisions about computation offloading and energy allocation, as shown by [141]. The E2E_DRL method uses a neural network to approximate the Q-values and learns from the environment through trial and error, demonstrating the potential of deep learning in edge computing.

Moreover, the applications of edge intelligence, including IoT, smart cities, and industrial automation, rely heavily on machine learning and deep learning [142]. The use of machine learning and deep learning in edge computing can enable real-time processing and analysis of data, which is essential for applications that require low latency and high bandwidth. For example, edge-based video analytics can use machine learning and deep learning techniques for real-time video processing and analysis, as discussed by [143].

The concept of learning-driven communication, which integrates wireless communication and machine learning to enable fast intelligence acquisition from distributed data, is also critical in edge computing [144]. This concept highlights the importance of machine learning and deep learning in edge computing, as they enable the development of intelligent edge devices and applications that can process and analyze data in real-time.

In addition, model compression, knowledge distillation, and multi-task learning are essential techniques in edge computing, as they enable the efficient processing and analysis of data at the edge [145]. These techniques can be used to optimize machine learning and deep learning models, reducing their complexity and improving their performance in edge computing environments.

The survey paper [146] compares the performance of different CI techniques in addressing critical issues in CC and EC, highlighting the strengths and weaknesses of each technique. The results of the survey paper show that LBS is the most commonly used CI technique in CC and EC, followed by EAs and SIAs. This highlights the importance of machine learning and deep learning in edge computing, as they enable the development of intelligent edge devices and applications that can process and analyze data in real-time.

Furthermore, the use of machine learning and deep learning in edge computing can also improve detection accuracy, reduced false alarm rates, and enhanced privacy preservation, as discussed in [147]. The use of deep learning (DL) approach for caching decision generation, model-based optimization module, stochastic gradient descent (SGD) algorithm, as proposed in [148], can also optimize edge computing performance. Additionally, the use of Deep Neural Networks (DNNs) in edge computing, DNN partitioning across multiple users, adaptive resource allocation to minimize latency, as discussed in [149], can further enhance edge computing efficiency.

The 6G White Paper [28] also discusses the use of machine learning and deep learning in edge computing, highlighting the need for real-time and distributed training of ML/AI algorithms. Moreover, the application of model-free deep reinforcement learning to provide a low-complexity sub-optimal solution to the MINLP problem in computation offloading and edge-cloud collaboration, as proposed in [137], can optimize edge computing performance. The use of machine learning and deep learning in MEC, for computation offloading and resource management, using game theory and reinforcement learning solutions, as discussed in [129], can also improve edge computing efficiency.

5.3 Edge Intelligence Frameworks and Applications

Edge intelligence frameworks and applications are crucial components of edge computing, enabling the efficient processing and analysis of data at the edge of the network . According to [138], the survey discusses several applications of Edge Intelligence, including wireless networking, service placement, computation offloading, and smart cities. The application of edge intelligence in these domains can improve system efficiency, reduce latency, and enhance security and privacy.

The existing edge intelligence frameworks and applications include the proposed Learning-Driven Communication (LDC) method, which can be applied in federated learning systems [144]. The LDC method integrates wireless communication and machine learning to enable fast intelligence acquisition from distributed data. Additionally, the use of various development boards, such as Google Edge TPU, Sophon Edge, Sipeed Maixduino, Ultra96, Nvidia Jetson Nano, and BeagleBone AI, can enable Edge-AI processing [143]. These development boards can be used to deploy machine learning models at the edge, reducing latency and improving system efficiency.

The application of FL-DRL method in edge computing can enable each IoT device to make decisions independently according to its own dynamic environment, while reducing transmission consumption and enhancing data privacy protection [118]. The FL-DRL method uses federated learning to train machine learning models in a distributed manner, reducing the need for data transmission to the cloud. Furthermore, the use of deep learning approach for caching decision generation, model-based optimization module, stochastic gradient descent (SGD) algorithm, can optimize edge computing performance [148].

The IAO method, which is an iterative algorithm for optimizing resource allocation, can prioritize resource allocation based on latency, ensuring that the user experiencing the highest latency receives sufficient resources [149]. This method can be used to optimize resource allocation in edge computing scenarios, improving system efficiency and reducing latency. The network-based compute reuse architecture, which uses a Reuse Store Table to store previously executed tasks, can also optimize edge computing performance [116]. This architecture can reduce the need for redundant computations, improving system efficiency and reducing latency.

In addition, the infrastructure types, including cloud, fog, and edge computing, can be used to deploy edge intelligence frameworks and applications [140]. The application architectures, including monolithic and microservices, can also be used to deploy edge intelligence frameworks and applications. The IoT use cases, including static and dynamic, can be used to deploy edge intelligence frameworks and applications. The methods, including heuristics, meta-heuristics, machine learning, and deep reinforcement learning, can be used to optimize edge computing performance. The resource management strategies can also be used to optimize edge computing performance.

5.4 Benefits and Challenges of Edge Intelligence

Edge intelligence has numerous benefits, including optimizing energy use, strengthening security, and integrating with next-generation networks like 6G [140]. The benefits of edge intelligence also include improved accuracy and efficiency of edge computing applications [145]. According to [10], the benefits of edge intelligence include support for real-time applications and increased bandwidth. However, edge intelligence also faces challenges, such as security concerns and the need for standardization [10]. The high computational demand of DNN models and data discrepancy in real-world settings are also significant challenges in edge intelligence [145]. Furthermore, the challenges and limitations of using AI algorithms in MEC security and privacy, including the need for large amounts of data and computational resources, must be addressed [147]. On the other hand, the proposed method in [128] has the advantage of leveraging the helper node's computation and communication resources to assist the user node, thereby improving the MEC performance. Overall, edge intelligence has the potential to enable a wide range of applications, from smart cities and industrial automation to healthcare and transportation, but its benefits and challenges must be carefully considered to ensure efficient and effective deployment.

5.5 Future Directions for Edge Intelligence Research

Edge intelligence is a rapidly evolving field, and there are several future research directions that need to be explored to fully harness its potential . According to [138], the survey suggests several future research directions in Edge Intelligence, including the development of novel algorithms and frameworks, the investigation of new applications and use cases, and the addressing of the challenges and opportunities in this field. The development of dedicated edge intelligence platforms and frameworks is also a key future direction, as highlighted by [142].

Moreover, the exploration of new applications and domains, such as smart cities, industrial automation, and healthcare, is essential for edge intelligence research [142]. The investigation of new technologies and techniques, such as machine learning, deep learning, and reinforcement learning, is also critical for edge intelligence research [122]. The use of machine learning techniques to improve the performance of computation offloading algorithms is a promising area of research, as noted by [122].

In addition, the development of hybrid CI techniques, the application of CI techniques to address security and privacy issues, and the investigation of the potential of CI techniques in emerging areas such as edge intelligence and blockchain-assisted CC and EC are future research trends in CI techniques for CC and EC [146]. The integration of edge intelligence with next-generation networks like 6G is also a key future direction, as highlighted by [140].

Furthermore, the development of advanced control algorithms, predictive mobility management techniques, and integration of blockchain and distributed ledger technologies are future research

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directions for edge intelligence in integrated terrestrial and non-terrestrial networks [96]. The optimization of energy use, strengthening of security, and integration with next-generation networks like 6G are also critical future directions for edge intelligence research [140].

6 IoT Edge and Applications

6.1 Introduction to IoT Edge and Applications

The concept of IoT edge and its applications is a rapidly growing field, with numerous applications in smart cities, industrial automation, and healthcare, as discussed in [16, 114, 150]. The application of edge computing in IoT scenarios has several benefits, including improved system efficiency, reduced latency, and enhanced security and privacy [16]. The Internet of Things (IoT) has emerged to digitalise our daily tasks in various systems, for example, smart homes, smart cities, smart factories, smart grids, and smart healthcare [114, 150]. The application of edge computing in IoT scenarios is crucial for enabling real-time processing and analysis of data, leveraging its integrating role in advancing state-of-the-art technologies such as 5G, artificial intelligence, and augmented reality, and has the potential to support a wide range of IoT applications, including smart homes and cities, industrial automation, healthcare, manufacturing, agriculture, and transportation, by overcoming technology restrictions such as data processing, storage, and transmission, and addressing time-sensitive applications through its integration with cloud and fog computing architectures [114, 150].

Moreover, the introduction to IoT edge and its applications is essential in understanding the potential of edge computing in IoT scenarios, as highlighted in [16, 114, 150]. The concept of IoT edge is also discussed in [16], which highlights the importance of edge computing in enabling real-time processing and analysis of data in IoT scenarios. The application of edge computing in IoT scenarios, including smart cities, industrial automation, and healthcare, is discussed in [16], which highlights the benefits and challenges of IoT edge, including improved system efficiency, reduced latency, and enhanced security and privacy, while the challenges include the need for standardization, security concerns, and the management of distributed and heterogeneous resources [16].

6.2 Existing IoT Edge Architectures and Applications

Existing IoT edge architectures and applications are diverse and include various components and functionalities, such as smart transportation, smart health-care, and smart grids [1, 123, 106]. According to [1], existing IoT edge architectures and applications have the potential to support a wide range of IoT applications, from smart homes and cities to industrial automation and healthcare. The proposed system model in [123] combines mobile edge computing and spectrum sharing for 5G communications, which can be considered as an existing IoT edge architecture. The integration of edge computing in IoT scenarios can significantly enhance system efficiency, reduce latency, and improve security and privacy by leveraging its proximity to IoT devices, enabling real-time data processing, and minimizing the need for cloud-based computations, thereby supporting applications such as smart cities, healthcare, and industrial automation, while also addressing challenges related to data transmission, energy efficiency, and quality of life features [17, 139, 151, 150, 114].

The hierarchical structure of IoT edge architectures is further illustrated in Figure 6, which depicts the relationships between smart cities, edge computing, and IoT applications, and is supported by references to relevant papers [1, 106]. This structure highlights the potential of existing IoT edge architectures and applications in supporting a wide range of IoT applications. Moreover, existing IoT edge architectures and applications can be used to support various IoT devices and applications, including sensors, actuators, and smart devices [1, 106]. The evaluation setting of IoTSim-Edge consists of three case studies: healthcare system, smart building, and capacity planning for roadside units, which are examples of IoT edge architectures and applications [106]. These case studies demonstrate the potential of existing IoT edge architectures and applications in supporting a wide range of IoT applications, from smart homes and cities to industrial automation and healthcare.

6.3 Smart Cities and Industrial Automation

Edge computing has numerous applications in smart cities and industrial automation, enabling the efficient processing and analysis of data in real-time, as discussed in [140, 150, 152, 138, 153]. According to [140], edge computing can be applied in smart cities to improve the quality of life for

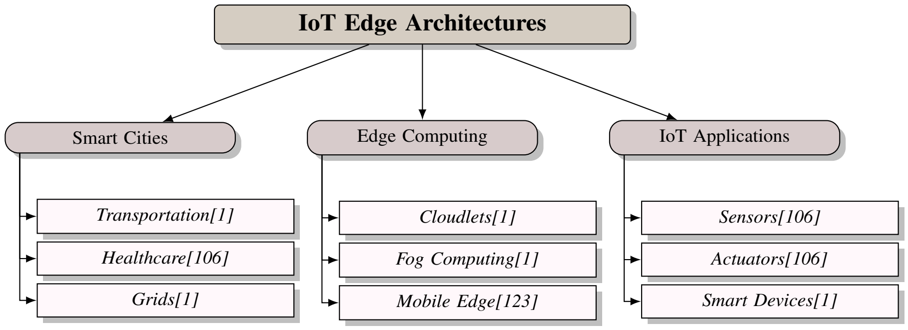

Figure 6: This figure shows the hierarchical structure of IoT edge architectures, including smart cities, edge computing, and IoT applications, with references to relevant papers.

citizens, by providing real-time processing and analysis of data from various sensors and devices. The use of edge computing in smart cities can also enable the development of intelligent transportation systems, smart grids, and smart buildings, making cities more efficient, sustainable, and livable [140]. Furthermore, edge computing can enable the development of intelligent industrial automation systems that can adapt to changing conditions and make decisions in real-time, leveraging advances in artificial intelligence (AI) and machine learning (ML) to improve the overall efficiency and productivity of manufacturing processes, while also supporting the integration of emerging technologies such as 5G networks, Internet of Things (IoT), and augmented reality, and enabling new applications such as immersive video conferencing, autonomous vehicles, and high-precision manufacturing, by providing instantaneous data processing and analytics at the edge of the network, rather than relying on centralized cloud computing [150, 152, 138, 153].

6.4 Healthcare and Other IoT Applications

Edge computing has numerous applications in healthcare and other IoT domains, enabling the efficient processing and analysis of data in real-time, as discussed in [140, 121, 154]. According to [140], edge computing can be applied in healthcare to improve the quality of patient care, by providing real-time processing and analysis of data from various medical devices and sensors. The use of edge computing in healthcare can also enable the development of intelligent healthcare systems, such as telemedicine, remote patient monitoring, and personalized medicine, making healthcare more efficient, sustainable, and effective [140]. Furthermore, edge computing can leverage the power of machine learning and deep learning algorithms to predict patient outcomes, detect anomalies, and optimize treatment plans, while also improving data transmission times and security through its low-latency and high-bandwidth capabilities, enabled by 5G communication, and can facilitate the deployment of intelligent edge services, enabling dynamic and adaptive edge maintenance and management, and paving the way for more pervasive and fine-grained intelligence in healthcare applications [17, 154].

6.5 Challenges and Future Directions

The IoT edge and applications face several challenges and future directions that need to be addressed to ensure the efficient and secure operation of these systems, as discussed in [38]. One of the primary challenges is the security and privacy of EC-assisted IoT, which requires lightweight security protocols, trust management frameworks, and privacy-preserving data analytics schemes [38]. The need for secure and private data processing and analysis is critical in IoT edge and applications, as the data generated by these systems is often sensitive and requires protection [38]. The use of blockchain technology and other distributed ledger technologies can provide a secure and transparent way to manage data in IoT edge and applications [38]. Additionally, the IoT edge and applications require trust management frameworks that can ensure the integrity and authenticity of the data generated by these systems [38]. The use of federated learning and other machine learning techniques can provide a privacy-preserving way to analyze data in IoT edge and applications [38].

7 Challenges and Open Research Issues

Edge computing, with its potential to revolutionize data processing and analysis, faces numerous challenges and open research issues, including security, scalability, and privacy concerns, as highlighted in [86, 87, 43, 38, 17, 147, 155]. The implementation of robust security protocols and mechanisms, such as encryption, access control, and blockchain technology, is crucial in providing a secure and reliable environment for various applications, including Internet of Things (IoT), smart cities, industrial automation, and healthcare, which can benefit from edge computing's low-latency and high-bandwidth capabilities, while addressing security and privacy concerns associated with the massive growth of IoT devices and data traffic, and leveraging the potential of artificial intelligence (AI) and edge computing to enhance data analysis, security, and efficiency [43, 38, 17, 147, 155].

7.1 Security Challenges

Security challenges in edge computing include ensuring data and computation privacy and security at the edge, and preventing unauthorized access to edge resources, as discussed in [86, 87, 10, 104, 53]. The development of new security protocols and architectures is crucial to ensure the trustworthiness, scalability, and dependability of edge computing systems, which are vulnerable to failures, security breaches, and privacy violations due to their distributed nature, limited resources, and proximity to data sources, and therefore require innovative solutions to address decentralized management, multi-tenancy, and interoperability challenges [10, 104, 53]. As illustrated in Figure 7, the security challenges in edge computing can be understood through a hierarchical structure, comprising data privacy, computation security, and system security, which highlights the complexity and interconnectedness of these challenges. Artificial intelligence and machine learning techniques can enhance edge computing system security by detecting and preventing cyber threats in real-time, as noted in [43, 38].

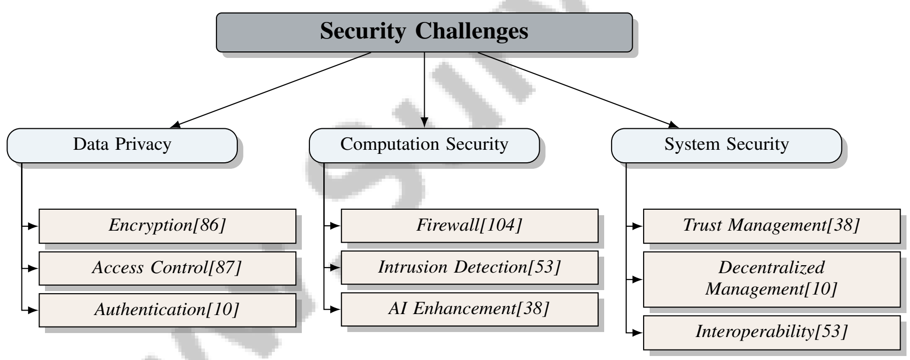

Figure 7: This figure shows the hierarchical structure of security challenges in edge computing, including data privacy, computation security, and system security.

7.2 Scalability and Performance Challenges

Edge computing faces numerous scalability and performance challenges, including efficient resource allocation and provisioning, and potential limitations such as limited resources and high energy consumption, as discussed in [135, 46, 128, 52, 130]. The complexity of real-time minimization problems, which may require significant computational resources, is a significant challenge, as noted by [128]. Scalability challenges, such as the complexity of the LMA, which increases with the number of layers and nodes in the HetMEC network, are emphasized by [52]. The limitation of proposed approaches, which require careful selection of control parameters to balance execution cost and battery energy level, is a scalability and performance challenge, as discussed by [130].

7.3 Privacy and Trust Challenges

Edge computing faces numerous privacy and trust challenges, including the need for distributed training schemes that protect data privacy and reduce transmission consumption, as discussed in [118, 115, 100, 101]. The provision of secure and private data processing, and the provision of secure and reliable services, are critical in edge computing, particularly in Mobile Edge Computing (MEC), as emphasized by [100]. Considering the specific features of edge paradigms, such as context awareness and interaction with mobile clients, during the development of security-aware software systems is also important, as noted by [101].

7.4 Open Research Issues and Future Directions

Edge computing is a rapidly evolving field, and several open research issues and future directions need to be explored to fully harness its potential, as discussed in [114, 97, 89, 105, 115, 140, 142, 90, 1, 113, 92, 156, 146, 10, 99, 126, 95, 48, 129, 111]. Future work on IoT integration with cloud, fog, and edge computing architectures should focus on addressing challenges and limitations, including standardization, security concerns, and limited resources, and exploring new applications and services that can be supported by these architectures, as suggested by [97, 89]. Developing a sustainable development roadmap for Edge Computing, considering technical, economic, and social dimensions of sustainability, and addressing gaps in current research, including the need for a comprehensive understanding of Edge Computing initiatives, is also a future direction, as noted by [89]. Open research issues in edge computing include developing standards, benchmarks, and marketplaces, frameworks and languages, lightweight libraries and algorithms, micro operating systems, and virtualisation, and the need for academia and industry to collaborate in driving research in this area, as emphasized by [105, 115]. Future research directions also include exploring emerging technologies and creative ideas to improve offloading performance, such as using Big Data techniques, 5G networks, and IoT applications, as suggested by [115]. Optimizing energy use, strengthening security, and integrating with next-generation networks like 6G are critical future directions for edge computing, as noted by [140]. The need for efficient processing and analysis of edge data, and the potential for edge intelligence to enable real-time decision-making in various domains, are also open research issues and future directions for edge computing, as discussed in [142]. Potential directions for future research include developing more effective architecture approaches and features, investigating new offloading and composition techniques, and applying Microservices in emerging Edge Computing scenarios such as IoT and 5G networks, as suggested by [90]. Developing more advanced edge computing architectures and algorithms is also an open research issue and future direction, as noted by [1]. Additionally, the need for more efficient resource management techniques, better support for heterogeneous devices, and improved security mechanisms in fog/edge computing is highlighted by [113]. The need for further study on the architecture, applications, and relationship of fog computing with other computing paradigms is also emphasized by [92]. The need for efficient task allocation in ephemeral edge computing is noted by [156]. The gaps in current research on CI techniques in CC and EC, including the need for more studies on the application of CI techniques to address security and privacy issues, the development of hybrid CI techniques, and the investigation of the potential of CI techniques in emerging areas such as edge intelligence and blockchain-assisted CC and EC, are identified by [146]. The future directions for edge computing, including exploring decentralized security mechanisms, developing standardized APIs, and investigating efficient resource management techniques, are suggested by [10]. The possibility of allowing EdgeDis to resist malicious nodes and cyberattacks, and the use of coded settings and other recovery mechanisms to further improve its performance, are proposed by [99]. The open research issues and future directions for edge computing, including the investigation of the application of Blockchain technology in MEC, the development of new offloading schemes, and the exploration of the tradeoff between latency and mining security, are identified by [126]. The need for more comprehensive and systematic evaluations of fog and edge computing systems, and the lack of standardized benchmarks and evaluation metrics, are noted by [95]. The open research issues and future directions for MEC-enabled Metaverse, including the need for further investigation into the design of an edge-cloud-enabled hierarchical Metaverse system and the development of new technologies and applications, are highlighted by [48]. The need for more advanced mathematical models, and the consideration of practical aspects such as heterogeneities in edge servers and fairness among servers, are emphasized by [129]. The future directions for research, including the development of more advanced AI and ML techniques, the integration of fog and edge computing with other technologies, such as blockchain and IoT, and the exploration of new applications and use cases, are suggested by [111].

8 Conclusion

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